

Recurrence Relation

① Iteration Method

Problem 1

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$T(1) = 1 \rightarrow \text{Base Case}$$

Solution:-

$$T(n) = 2T\left(\frac{n}{2}\right) + n \quad \text{--- ①}$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{2 \times 2}\right) + \frac{n}{2}$$

$$T\left(\frac{n}{2}\right) = 2T\left[\frac{n}{2^2}\right] + \frac{n}{2} \quad \text{--- ②}$$

$$T\left(\frac{n}{2^2}\right) = 2T\left(\frac{n}{2^2 \times 2}\right) + \frac{n}{2^2}$$

$$= 2T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} \quad \text{--- ③}$$

Substitute eqn ② and ③ in eqn ①

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

$$= 2\left[2T\left(\frac{n}{2^2}\right) + \frac{n}{2}\right] + n$$

$$= 2^2\left[T\left(\frac{n}{2^2}\right) + \cancel{2} \cdot \frac{n}{\cancel{2}}\right] + n$$

$$= 2^2 T\left(\frac{n}{2^2}\right) + 2n$$

$$= 2^2 \left[2 T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} \right] + 2n$$

$$= 2^3 \left[T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} \right] + 2n$$

$$T(n) = 2^3 T\left(\frac{n}{2^3}\right) + 3n \quad \text{--- (4)}$$

Now, generalize eqn (4)

$$T(n) = 2^i T\left(\frac{n}{2^i}\right) + in \quad \text{--- (5)}$$

Apply i value in eqn (5)

$$T(n) = n \cdot T(1) + \log_2 n \cdot n \quad \text{--- (6)}$$

Since $T(1) = 1$

So, eqn (6) becomes

$$= n \cdot 1 + n \cdot \log_2 n$$

$$= n + n \log_2 n$$

Take the higher order term, so the time complexity is

$$\boxed{O(n \log_2 n)}$$

Consider

$$\frac{n}{2^i} = 1$$

$$\boxed{n = 2^i}$$

Apply log on both sides

$$\log_2 n = \log_2 2^i$$

$$\log_2 n = i \log_2 2$$

$$\boxed{\log_2 n = i}$$